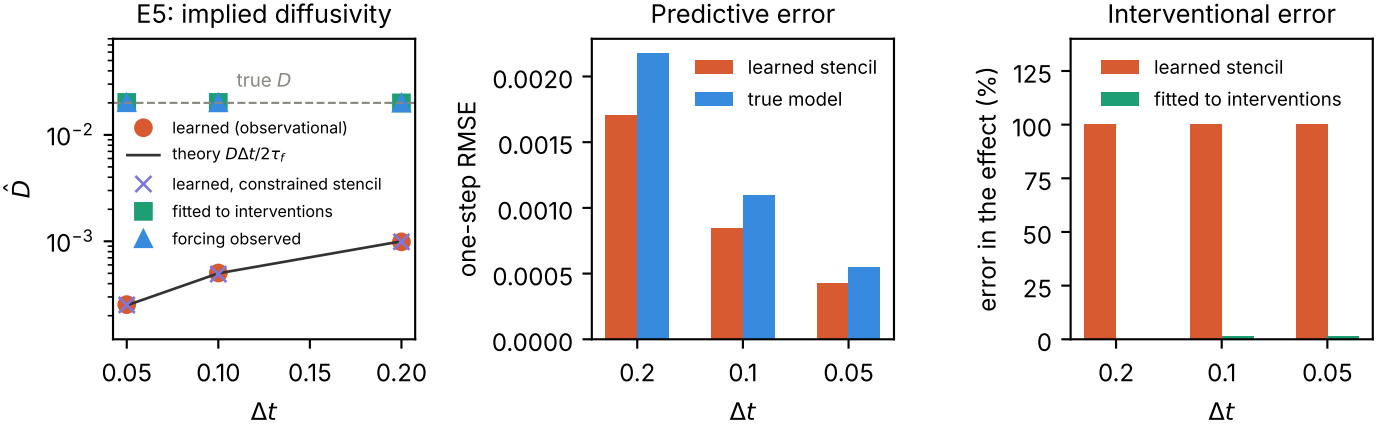
E5: implied diffusivity
true D
learned (observational)
theory $D\Delta t/2\tau_f$
learned, constrained stencil
fitted to interventions
forcing observed
$\hat{D}$
10^{-2}
10^{-3}
0.05
0.10
0.15
0.20
Δt
Predictive error
learned stencil
true model
one-step RMSE
0.0020
0.0015
0.0010
0.0005
0.0000
0.2
0.1
0.05
Δt
Interventional error
learned stencil
fitted to interventions
error in the effect (%)
125
100
75
50
25
0
0.2
0.1
0.05
Δt